

d.RADIUS & TACACS+ : TACACS+ & RADIUS Features, RADIUS Authentication Process, TACACS+ Use Cases, RADIUS vs. TACACS+

Task 1.

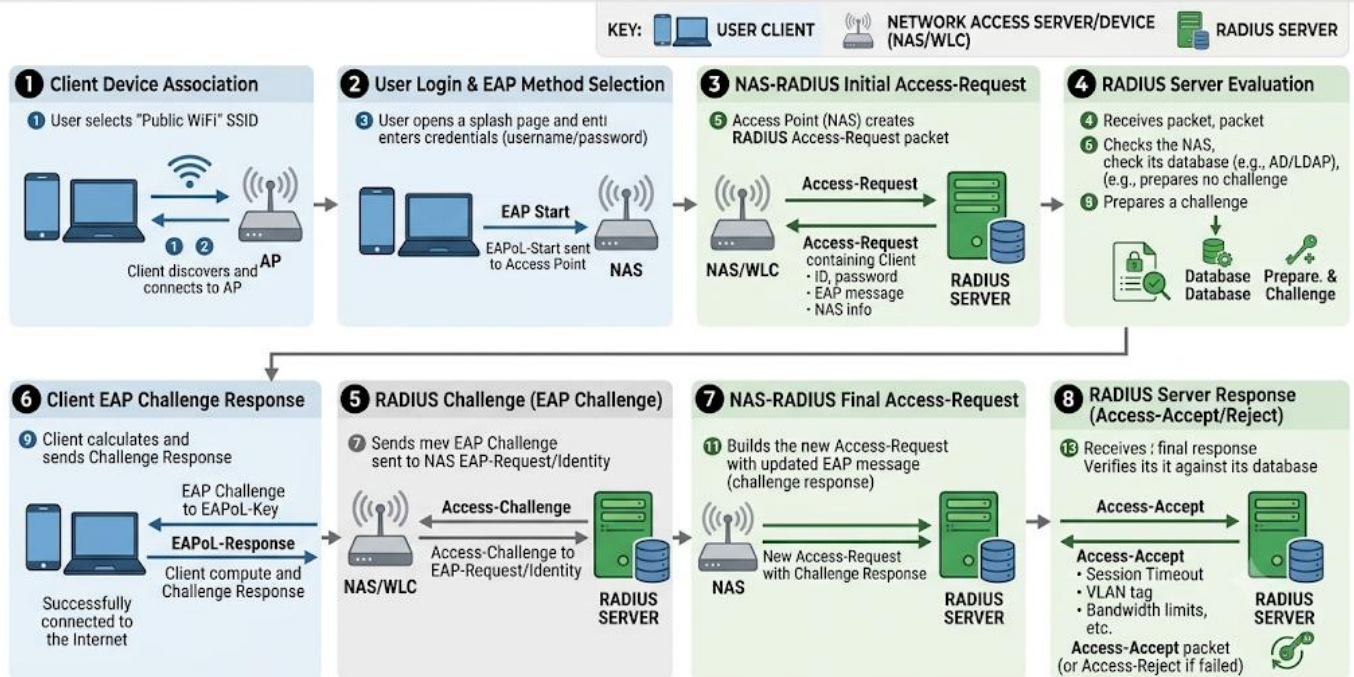
Create a comparison table listing at least 4 key features of RADIUS and TACACS+ side by side, highlighting which protocol supports each feature.

Feature	RADIUS	TACACS+
Authentication	Supported	Supported
Authorization	Supported, but more limited	Strong support
Accounting	Supported	Supported
Encryption	Encrypts the user's password; other packet information is not fully encrypted	Encrypts the entire packet body
Transport protocol	UDP	TCP
Default ports	UDP 1812 (authentication), UDP 1813 (accounting)	TCP 49
Command-level authorization	Limited	Supported
Common use	WiFi, VPN, network access	Cisco/network-device administration

Task 2.

Draw a step-by-step flow diagram showing the RADIUS authentication process for a user logging in to a public WiFi hotspot, labeling each communication step between the client, network device, and RADIUS server.

STEP-BY-STEP FLOW: PUBLIC WIFI HOTSPOT RADIUS AUTHENTICATION PROCESS



Task 3.

List 3 real-world use cases where TACACS+ is preferred over RADIUS, and briefly explain why TACACS+ is better suited for each scenario.

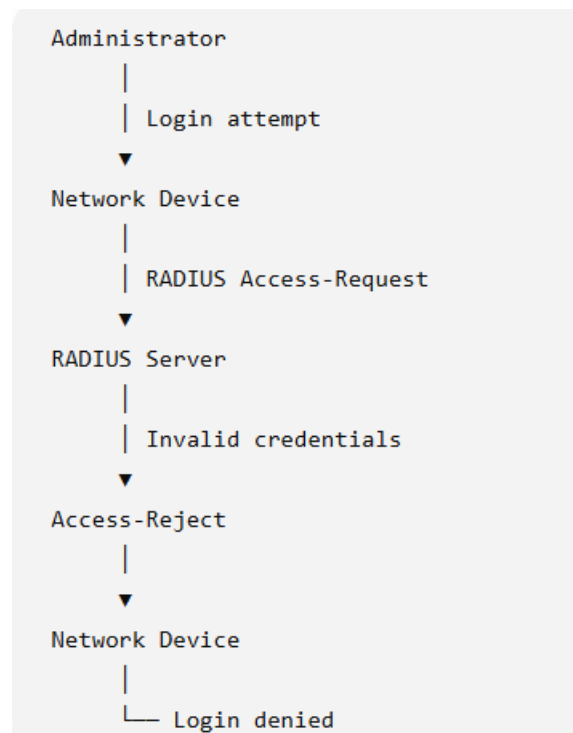
Scenario	Why TACACS+ is suitable
1. Cisco router/switch administration	TACACS+ provides detailed authorization of administrator commands, allowing different administrators to have different command permissions.
2. Network Operations Center (NOC)	Network administrators can be authenticated centrally and their individual commands can be authorized and audited.
3. Enterprise network management	TACACS+ separates authentication, authorization, and accounting, making it useful when organizations need detailed control over administrator access to network devices.

Task 4.

Given a scenario where a user tries to log into a network device but fails authentication, describe how RADIUS and TACACS+ would handle the logging and response differently.

Hint: Focus on how each protocol manages accounting and command authorization.

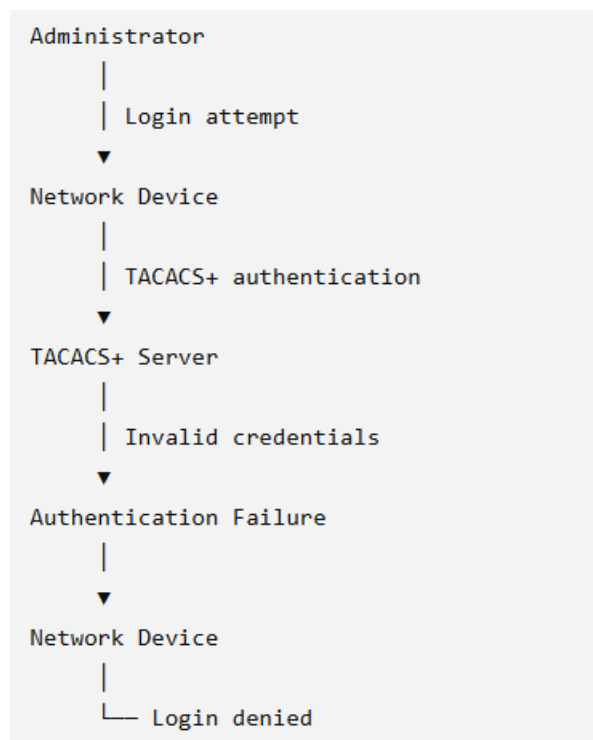
Suppose an administrator enters an incorrect password while trying to log into a network device.



RADIUS can record authentication/accounting information such as session activity. However, RADIUS is primarily designed around network access, such as WiFi and VPN authentication.

TACACS+

TACACS+ is particularly useful for network-device administration because it can separately handle authentication, authorization, and accounting. It can also provide command-level authorization and logging.



Key difference: If authentication succeeds, TACACS+ can additionally determine whether the administrator is permitted to execute a particular command. RADIUS has much less granular command authorization capability.

Task 5.

Use ChatGPT or another AI tool to generate a sample RADIUS configuration snippet for a Cisco router, then review the output and list any changes you would make to adapt it for a college campus WiFi setup.

Commands

```
Router(config)# aaa new-model
```

```
Router(config)# radius server CAMPUS-RADIUS
```

```
Router(config-radius-server)# address ipv4 192.168.100.10 auth-port 1812 acct-port 1813
```

```
Router(config-radius-server)# key CampusSecret123
```

```
Router(config-radius-server)# exit
```

```
Router(config)# do wr
```

Router(config)# aaa authentication login default group radius local

Router(config)# aaa authorization exec default group radius local

Router(config)# do wr

Configuration item	Campus adaptation
RADIUS server IP	Replace 192.168.100.10 with the actual campus RADIUS server IP.
Shared secret	Use a strong, unique secret shared between the network device and RADIUS server.
User database	Connect RADIUS to the college's existing student/staff directory or identity system.
WiFi device	Configure each campus AP/WLAN controller as a RADIUS client.
Accounting	Enable RADIUS accounting to record student/staff WiFi sessions.
VLAN assignment	Configure policies so students, faculty, and guests can receive appropriate VLAN/network access.
Security	Use WPA2/WPA3-Enterprise with 802.1X rather than a single shared WiFi password.
Redundancy	Add a secondary RADIUS server so authentication can continue if the primary server is unavailable.